PIMPRI CHINCHWAD EDUCATION TRUST'S

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

Department of Applied Science and Humanities (AS&H)

Report of FA-2 activity

Subject: BSH23OE04: Open Elective: Statistical Data Analysis using R

Academic Year 2026-27

Semester I

Submitted by

Sr. No.	Name of the student	PRN number	Branch
1.			
2.			
3.			

Submitted to: Dr. Rajesh G Talekar

Date: _____ Sign: _____

Project Title

Statistical Analysis of the Vietnam War Bombing Operations Data using R

Data used

THOR Vietnam War bombing operations database (Combat Archives, THOR project). File: THOR_Vietnam_Bombing_Operations.csv. The file is public research data on weapon-delivery records of the air campaign in South-East Asia.

Details of the data

The file contains 46,70,416 records (rows) and 47 columns covering mission dates from 1 June 1965 to 30 June 1975. One row is one weapon delivery event as reported by one source database (SEADAB, CACTA6669, CACTA6570 or SACCOACT); the column SOURCERECORD identifies the source. The columns are listed below together with the share of values missing in the full file (computed by a complete scan of the file, script scripts/00_profile_raw.R).

No.	Column	Type	Description	Missing
1	THOR_DATA_VIET_ID	Number	Unique record identifier in the THOR Vietnam database; checked to be unique (46,70,416 distinct values).	0%
2	COUNTRYFLYINGMISSION	Category	Country whose air force flew the mission (USA, South Vietnam, Laos, South Korea, Australia).	0.08%
3	MILSERVICE	Category	Military service of the flying unit: USAF, USN, VNAF, USMC, RLAF, KAF, RAAF, OTHER, USA.	0.07%
4	MSNDATE	Date	Mission date; stored as text in two formats (DD-MMM-YY and YYYYMMDD), parsed in cleaning.	0%
5	SOURCEID	Category	Identifier of the source document the record came from.	0%
6	SOURCERECORD	Category	Source database of the record: SEADAB, CACTA6669, CACTA6570, SACCOACT.	0%
7	VALID_AIRCRAFT_ROOT	Category	Validated aircraft type code (F-4, F-100, A-4, B-52, ...), 129 levels.	0%
8	TAKEOFFLOCATION	Category	Air base, carrier or other location the aircraft took off from.	0.11%
9	TGTLATDD_DDD_WGS84	Number	Target latitude, decimal degrees, WGS84.	24.20%
10	TGTLONDDD_DDD_WGS84	Number	Target longitude, decimal degrees, WGS84.	24.20%
11	TGTTYPE	Category	Text description of the target type (AREA\DEPOT, BUILDINGS, TRUCKS, ...), 268 levels.	39.19%
12	NUMWEAPONSDELIVERED	Number	Number of weapons delivered on the target in the record.	0%
13	TIMEONTARGET	Number	Time over target, encoded as HHMM.	0.57%
14	WEAPONTYPE	Category	Weapon type designation (e.g. MK-82, 750LB GP).	51.46%
15	WEAPONTYPECLASS	Category	Weapon class in the source file: completely empty in this file.	100%
16	WEAPONTYPEWEIGHT	Number	Weight of one weapon of this type, pounds.	0%
17	AIRCRAFT_ORIGINAL	Category	Aircraft type as written in the source document.	0.01%
18	AIRCRAFT_ROOT	Category	Aircraft type code without model suffix (F4, F100, A4, B52, ...).	0.01%
19	AIRFORCEGROUP	Category	Air force group code; almost completely missing, not used.	99.94%
20	AIRFORCESQDN	Category	Squadron; almost completely missing, not used.	99.94%
21	CALLSIGN	Category	Mission callsign.	70.66%
22	FLTHOURS	Number	Flight hours logged for the record.	0%
23	MFUNC	Category	Mission function code.	2.16%
24	MFUNC_DESC	Category	Mission function description (STRIKE, CLOSE AIR SUPPORT, ...), 95 levels.	2.24%
25	MISSIONID	Category	Mission identifier.	0.34%
26	NUMOFACFT	Number	Number of aircraft taking part in the record.	0%
27	OPERATIONSUPPORTED	Category	Name of the operation supported; text is very inconsistent (51,375 levels).	41.11%
28	PERIODOFDAY	Category	Period of day of the attack: D (day) or N (night); other codes are data-entry noise.	4.28%

29	UNIT	Category	Flying unit name.	0.01%
30	TGTCLOUDCOVER	Category	Cloud cover at the target: CLEAR, SCATTERED, BROKEN, OVERCAST, UNDERCAST, NOT OBS, IN CLOUDS, BETWEEN LAYERS.	52.25%
31	TGTCONTROL	Category	How the target was controlled/acquired (ABN FWD AIR CNTL, TPQ-10, ...).	46.05%
32	TGTCOUNTRY	Category	Country of the target: South Vietnam, Laos, North Vietnam, Cambodia, Thailand, Westpac Waters, UNKNOWN, PHILLIPINES (typo in source).	4.64%
33	TGTID	Category	Target identifier; almost completely missing, not used.	100%
34	TGTORIGCOORDS	Category	Target coordinates as recorded originally.	22.87%
35	TGTORIGCOORDSFORMAT	Category	Format of the original coordinates (7 levels).	23.40%
36	TGTWEATHER	Category	Coded weather at the target; free-text style coding, 16,235 levels.	55.51%
37	ADDITIONALINFO	Category	Additional free-text information about the record.	0%
38	GEOZONE	Category	Target geozone code (84 levels).	25.01%
39	ID	Number	Internal record id.	0%
40	MFUNC_DESC_CLASS	Category	Mission function class: KINETIC or NONKINETIC.	0%
41	NUMWEAPONSJETTISONED	Number	Number of weapons jettisoned; -1 is a missing code.	0%
42	NUMWEAPONSRETURNED	Number	Number of weapons returned to base; -1 is a missing code.	0%
43	RELEASEALTITUDE	Number	Release altitude; almost completely missing.	99.93%
44	RELEASEFLTSPEED	Number	Release flight speed; almost completely missing.	99.96%
45	RESULTSBDA	Category	Bomb damage assessment text; almost completely missing.	93.89%
46	TIMEOFFTARGET	Number	Time off target, encoded as HHMM.	0.57%
47	WEAPONSLOADEDWEIGHT	Number	Total weight of weapons loaded, pounds; -1 is a missing code; only filled in the SEADAB source.	0%

Table 1: Columns of the raw data file with the share of missing values in the full file.

Data preparation and sampling

The 47-column file mixes four source databases and contains records that did not deliver any weapon (for example jettison or return only sorties). The following steps were applied (script scripts/01_clean.R):

- Step 1 - Population. The full file was scanned and the 23,46,224 records with NUMWEAPONSDELIVERED > 0 (50.24% of the file) were kept. These are the records in which a weapon was actually delivered, which is the subject of the study.
- Step 2 - Sampling. A simple random sample without replacement of 20,000 records was drawn from those 23,46,224 records (set.seed(2026)). Working on a random sample keeps every run of the scripts fast while the sample is still large enough for every test in this report.
- Step 3 - Cleaning and derivation. The sample was cleaned and the analysis variables were derived; the result is clean_data.csv (20,000 records, 47 raw plus 17 derived columns) and a data-quality log (outputs/01_data_quality.txt).

Cleaning rules applied to the sample

- Mission dates: the 272 records stored in YYYYMMDD format were converted; all 20,000 dates parsed. The sample covers 1 October 1965 to 3 June 1975.
- NUMWEAPONSDELIVERED = 9999 (sentinel value): 2 records set to missing.
- WEAPONSLOADEDWEIGHT = -1 (missing code): 12,394 records set to missing; WEAPONSLOADEDWEIGHT = 0 (records that delivered non-ordnance such as flares or leaflets, so a weapon load does not apply): 1,600 records set to missing; values above 1,000,000 lb (implausible, transport aircraft row): 112 records set to missing.
- NUMOFACFT = 0 (impossible for a delivery record): 13 records set to missing.
- TIMEONTARGET: 8 records with impossible HHMM values set to missing; the valid values were converted to the hour of the day (TGT_HOUR).
- Coordinates outside the study box 0-30N, 90-130E: 2,456 records set to missing.
- TGTCLOUDCOVER 'NOT OBS' treated as missing (396 records); the rare categories 'IN CLOUDS' and 'BETWEEN LAYERS' grouped with 'Other'.
- TGTCOUNTRY 'PHILLIPINES' corrected to PHILIPPINES; 'WESTPAC WATERS' and 'UNKNOWN' grouped as 'Other'; MILSERVICE values OTHER and USA (118 records) grouped as 'Other'.
- Aircraft types were grouped into the six most used types (F-4, F-100, A-4, A-1, A-37, B-52) plus 'Other'; the weapon type was joined with the weapons glossary file to obtain WEAPON_CLASS (BOMB, ROCKET, GUN, MISSILE, SUPPORT); every weapon string matched the glossary.
- Derived variables: YEAR, MONTH from the mission date; PERIOD (Day/Night); COUNTRY, SERVICE, SOURCE_DB, AIRCRAFT_GROUP, CLOUD, WEAPON_CLASS as factors; WEAPONS_DELIVERED, LOAD_LB, AIRCRAFT_NO, TGT_HOUR, LAT, LON, FLIGHT_HOURS as numbers.

Missing values of the analysis variables in the 20,000-record sample

Variable	Missing	Share	Variable	Missing	Share
PERIOD	834	4.17%	LOAD_LB	14,106	70.53%
COUNTRY	349	1.75%	AIRCRAFT_NO	13	0.06%
AIRCRAFT_GROUP	1	0.00%	TGT_HOUR	8	0.04%
CLOUD	10,729	53.65%	LAT / LON	2,456	12.28%
WEAPON_CLASS	4,250	21.25%	WEAPONS_DELIVERED	2	0.01%
SERVICE / SOURCE_DB	0	0.00%	FLIGHT_HOURS	0	0.00%

Table 2: Missing values of the analysis variables after cleaning.

The weapon load (LOAD_LB) is recorded only in the SEADAB source, which is why 70.5% of the sample has no load; every problem that uses the load states this and analyses only the records where it is known. The cloud cover is missing for more than half of the records because ground reports often did not observe it.

Limitations of the data (stated once, valid for every problem)

- One record is one delivery event as reported by one source, not one mission. A single mission can produce several records, so records are not fully independent. The p-values in the tests are therefore slightly optimistic; the effect sizes are quoted for this reason.
- The four source databases overlap in time and purpose; the database merges them without deduplication flags, so double counting of unusual events cannot be fully excluded.
- Weapons loaded weight is recorded only in SEADAB; results about the load apply to that source only.
- The source data are historical reports of the war; recorded numbers (weapons, aircraft) are the numbers reported at the time.

Mapping of the course units to the problem statements

Unit	Topics in the syllabus	Covered by
Unit I	Data handling in R, descriptive statistics, data visualization	Problem statement 1
Unit II	Probability, random variables, standard distributions, central limit theorem	Problem statement 2
Unit III	Sampling, estimation, hypothesis testing (t-test, chi-square test), testing of means and proportions	Problem statements 3 and 4
Unit IV	Correlation, simple and multiple linear regression, prediction	Problem statement 5

Table 3: Coverage of the four units of the syllabus by the five problem statements.

Problem statement 1 (Unit I): Descriptive analysis of the weapon-delivery records

Problem statement

Describe how the weapon-delivery activity recorded in the data is distributed over time, target countries, aircraft types and weapon classes, and summarize the number of weapons delivered and the number of aircraft used per record with suitable descriptive statistics and graphs.

Introduction (purpose and approach)

The purpose is to build the first overall picture of the air campaign as recorded in the data: when it happened, where it happened, who flew it and what was delivered. The approach is classical descriptive statistics: frequency tables for the categorical variables, summary statistics (mean, median, quartiles, standard deviation, skewness, kurtosis) for the numerical variables and one graph per question, all produced with R and ggplot2.

Approach / algorithm with explanation

1. Load clean_data.csv (source scripts/common_setup.R).
2. Frequency tables of records by year, month, target country, aircraft group, period of day, source database and weapon class.
3. Summary statistics of WEAPONS_DELIVERED, AIRCRAFT_NO, LOAD_LB, TGT_HOUR, LAT and LON, including the median, quartiles, interquartile range, skewness and kurtosis, because the distributions are strongly skewed.
4. Graphs: bar chart of records per year; monthly time series by target country; histogram of weapons delivered; boxplots of weapons delivered by country; map of the target coordinates.
5. Read each number with attention to the heavy skew (comparison of mean and median).

Output with description

Frequency tables (from outputs/problem1.txt):

Year	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	1975
Records	300	1,954	2,337	3,352	2,470	3,532	2,161	2,642	910	256	86

Table 4: Number of weapon-delivery records per year in the sample.

Category	Records	Share	Category	Records	Share
South Vietnam	10,510	53.5%	F-4 Phantom	4,827	24.1%
North Vietnam	2,363	12.0%	F-100 Super Sabre	2,877	14.4%
Laos	5,736	29.2%	A-4 Skyhawk	1,841	9.2%
Cambodia	960	4.9%	A-1 Skyraider	2,067	10.3%
Thailand	82	0.4%	A-37 Dragonfly	1,515	7.6%
Day records	13,786	68.8%	B-52 Stratofortress	836	4.2%
Night records	5,380	26.8%	Other types	6,036	30.2%
BOMB class	12,362	61.8%	SEADAB source	7,606	38.0%
GUN class	535	2.7%	CACTA6669 source	9,944	49.7%

Table 5: Headline frequencies of the sample (shares are of the 20,000 records; weapon classes exclude the 4,250 records without a usable weapon type).

Variable	n	Mean	Median	Std. dev.	Q1	Q3	Max	Skewness
WEAPONS_DELIVERED	19,998	38.10	8	268.91	4	16	9,040	21.50
AIRCRAFT_NO	19,987	3.06	2	7.53	2	2	276	11.35
LOAD_LB (lb)	5,894	59,795.3	34,260	94,711.3	15,930	63,720	8,96,480	4.55
TGT_HOUR (hour of	19,992	8.98	8	6.23	4	13	23	0.52

day)								

Table 6: Summary statistics of the main numerical variables.

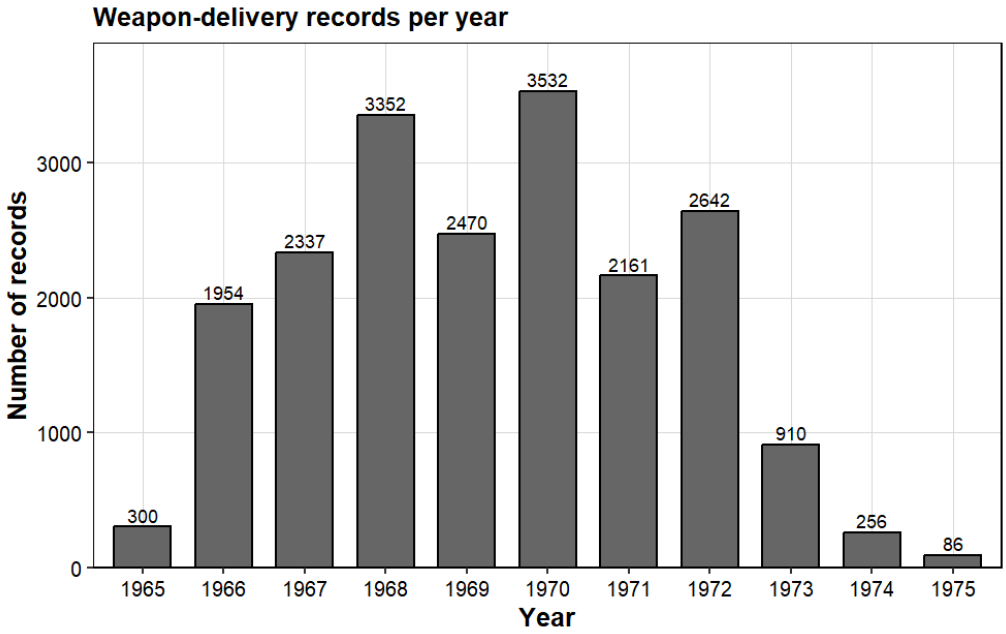


Figure 1.1: Weapon-delivery records per year. Activity builds up 1966-68, peaks in 1970 (3,532 records) and nearly stops after 1972 (86 records in 1975).

Inference: the war effort recorded in the data is concentrated in 1968-1971 (57.6% of all sample records); the sharp fall after 1972 matches the withdrawal of US forces and the end of large-scale bombing.

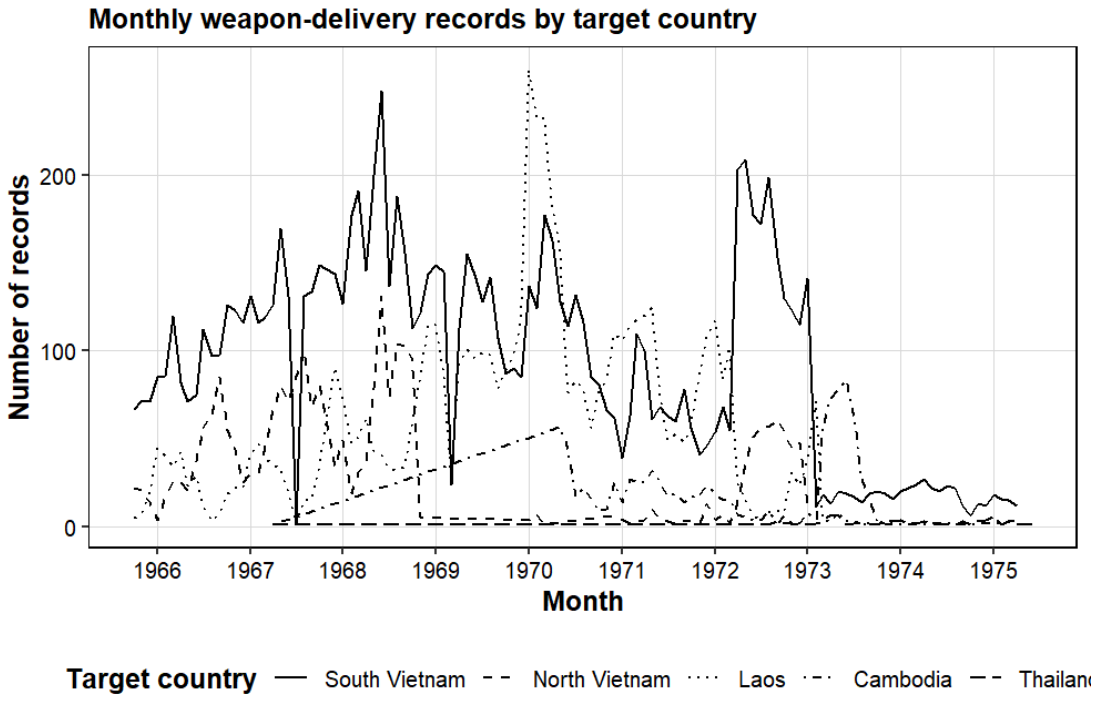


Figure 1.2: Monthly number of records by target country. South Vietnam dominates throughout; Laos is present steadily; the North Vietnam series appears mainly in 1972.

Inference: the monthly profile differs by country - South Vietnam is a continuous high-volume theatre, while North Vietnam shows concentrated campaigns (1967-68 and 1972), which is consistent with the historical record of the two bombing campaigns against the North.

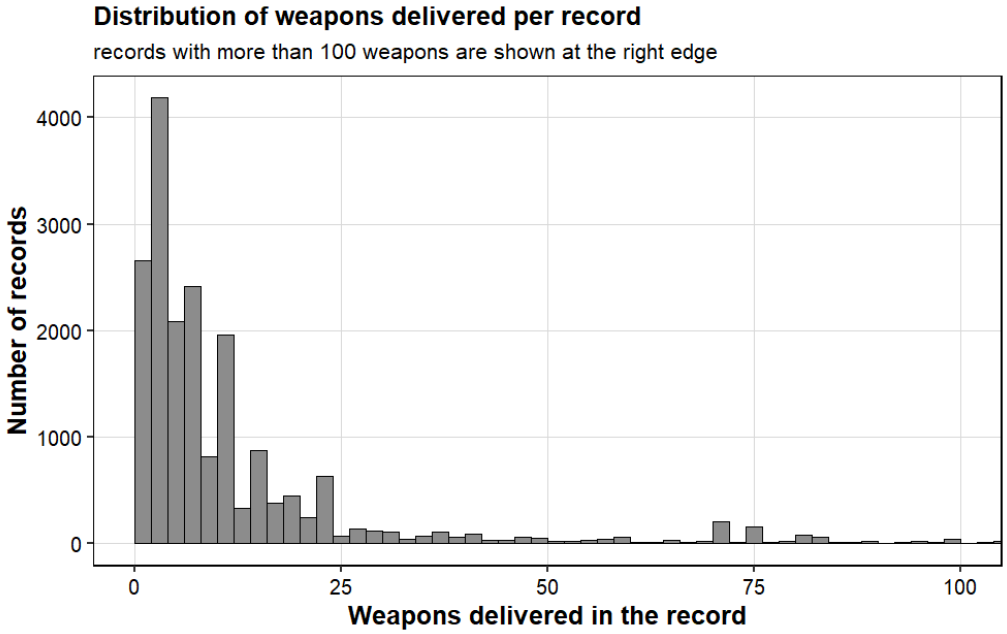


Figure 1.3: Number of weapons delivered per record (records above 100 weapons are shown at the right edge).

Inference: the distribution is extremely skewed: half of the records deliver 8 or fewer weapons, but 5.74% of the records deliver more than 100 (maximum 9,040). For this reason the median and the quartiles are reported next to the mean everywhere in this report.

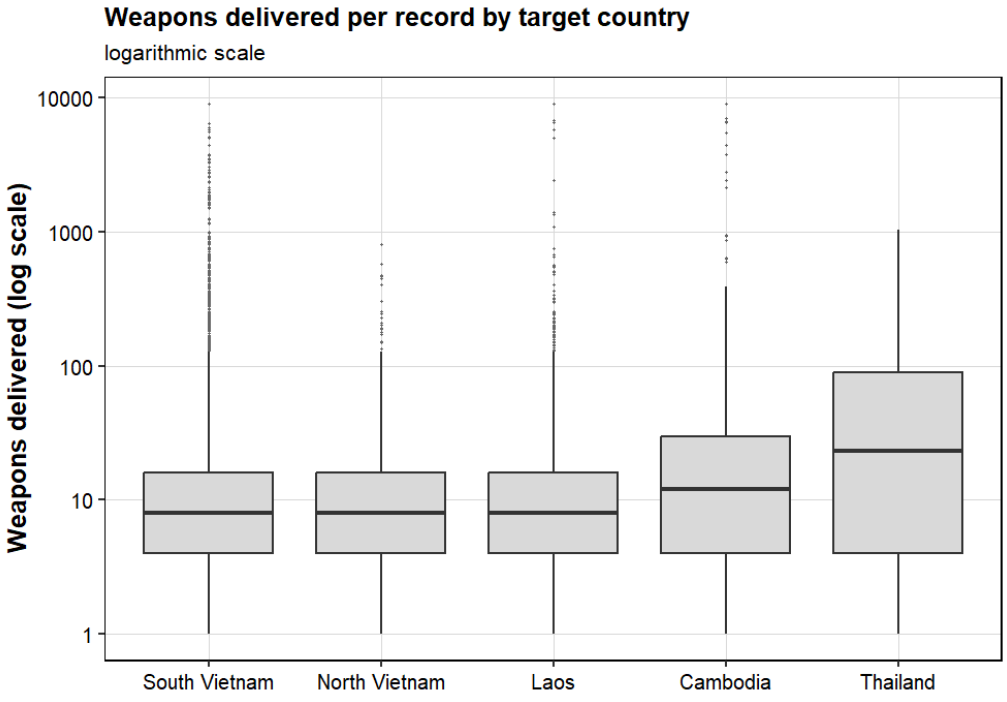


Figure 1.4: Weapons delivered per record by target country (boxplots).

Inference: the typical (median) record is small in every country (8 weapons in South Vietnam, North Vietnam and Laos, 12 in Cambodia); the means are much larger than the medians in every country because of the long right tail. Comparing countries by mean would be misleading, by median they are similar.

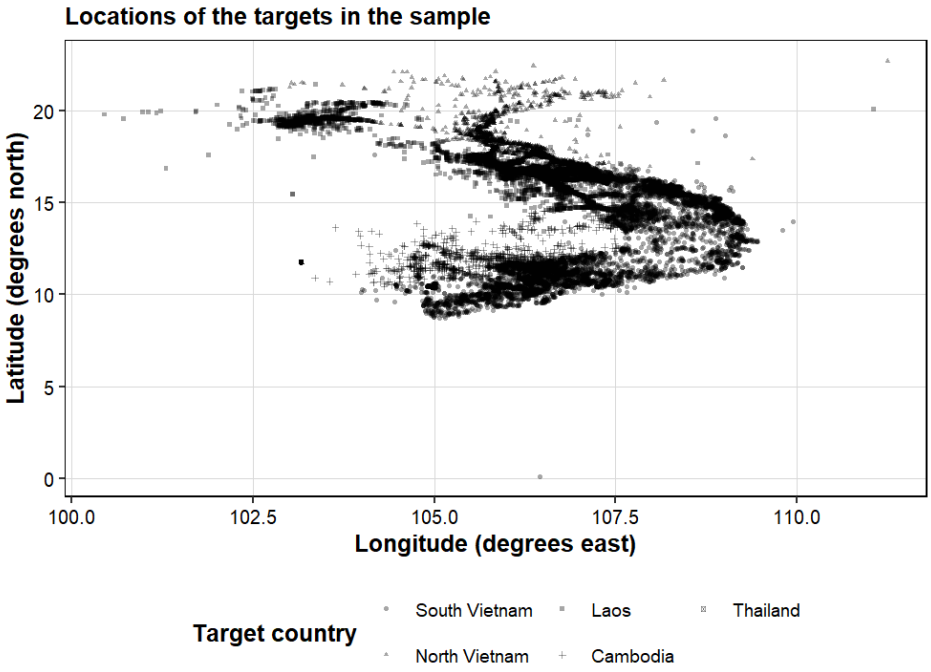


Figure 1.5: Target locations of the sampled records (longitude against latitude).

Inference: the target coordinates form the expected map of the theatre: a dense band along the South Vietnamese coast and the Mekong delta, the Laos panhandle, and clusters in the North. The coordinates agree with the recorded target country, so the geographic information is internally consistent.

Conclusion

The descriptive analysis shows a campaign concentrated in 1966-1972 with a clear peak in 1970; the main theatre is South Vietnam (53.5% of targets) followed by Laos (29.2%); the workhorse aircraft are the F-4 (24.1%), F-100 (14.4%) and A-1/A-4/A-37 families; bombing (BOMB class, 61.8% of records with a known class) is the dominant weapon; day records outnumber night records 13,786 to 5,380. Every numerical variable is strongly right-skewed, so medians and quartiles describe the typical record better than means.

Problem statement 2 (Unit II): A probability model for the weapon load

Problem statement

Find a probability distribution that describes the recorded weapon load of a delivery record. Compare the normal and the lognormal model, test the quality of the fit, use the model to compute probabilities, and check the central limit theorem for the mean load of 30 records.

Introduction (purpose and approach)

The weapon load (LOAD_LB) is recorded for the 5,894 sampled records of the SEADAB source (29.5% of the sample) and is the best continuous measure of the size of a delivery in the data. A probability model for it allows statements such as 'the chance that a record delivers more than 50,000 lb' to be computed. The approach is: estimate the parameters of the candidate models by maximum likelihood, test the goodness of fit with the chi-square test on equal-probability bins, and verify the central limit theorem by simulation.

Approach / algorithm with explanation

1. Take the 5,894 records with a recorded load; summarise them (mean, median, standard deviation, skewness, kurtosis, quartiles).
2. Fit the normal model: parameters equal to the sample mean and standard deviation (maximum likelihood for the normal).
3. Fit the lognormal model: take log10(load), fit a normal distribution to the logs, report the parameters in natural log units.
4. Chi-square goodness of fit for both models: split the fitted distribution into 8 equally likely bins, count the observed records in the same bins, compute the chi-square statistic with df = 7 - 2 estimated parameters = 5.
5. Compare observed and fitted quartiles and compute model probabilities; use boxplots of the load by aircraft group to investigate the misfit.
6. Central limit theorem: draw 20,000 samples of n = 30 records, form the sample means, compare their mean, standard deviation and shape with the theoretical values, and compute the probability that the sample mean exceeds 70,000 lb by theory and by simulation.

Output with description

Quantity	Value	Quantity	Value
n (records with load)	5,894	Median	34,260 lb
Mean	59,795.3 lb	Q1, Q3	15,930 / 63,720 lb
Std. deviation	94,711.3 lb	Minimum, maximum	60 / 8,96,480 lb
Skewness	4.554	Kurtosis	25.245

Table 7: The recorded weapon load (all 5,894 records come from the SEADAB source).

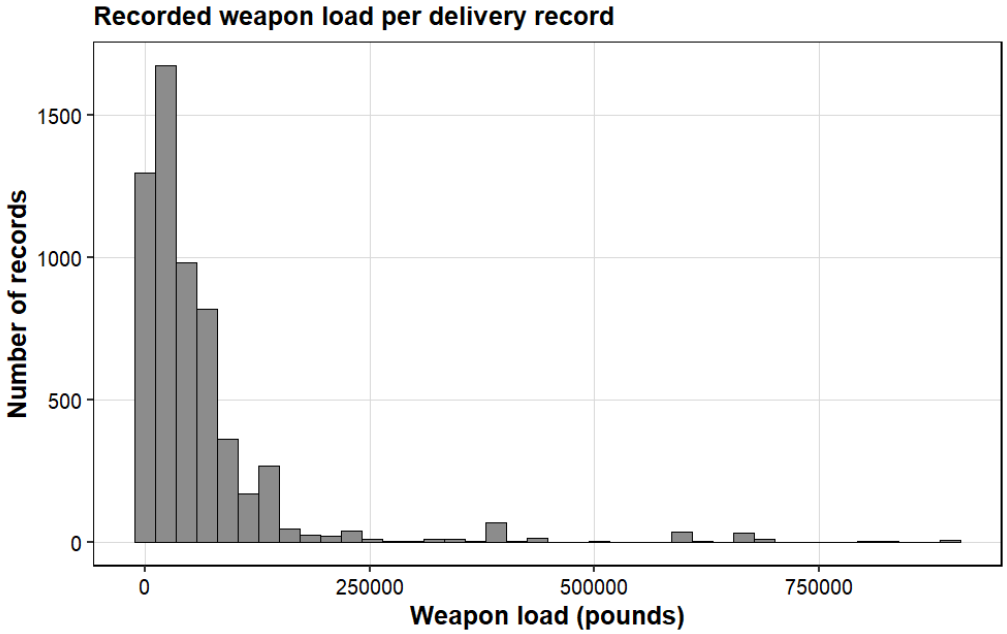


Figure 2.1: Histogram of the recorded weapon load; the strong right skew (skewness 4.55) makes a normal model implausible for the load itself.

Normal model: mean 59,795.3 lb, sd 94,703.2 lb. The chi-square statistic is 7,337.8 on 5 degrees of freedom ($p < 2.2e\text{-}16$), i.e. the model is rejected. The fitted normal puts 12.5% of the records in the lowest bin, but not a single record can be below the physical minimum of zero, so the lower tail is impossible.

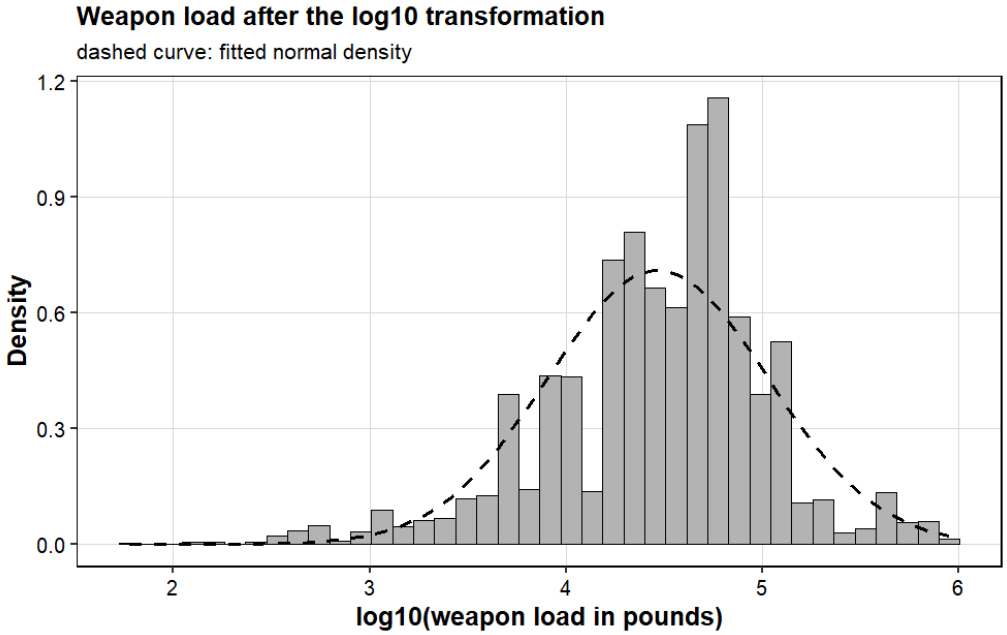


Figure 2.2: log10 of the weapon load with the fitted normal density; on the log scale the data are close to symmetric (skewness -0.64), which is the lognormal idea.

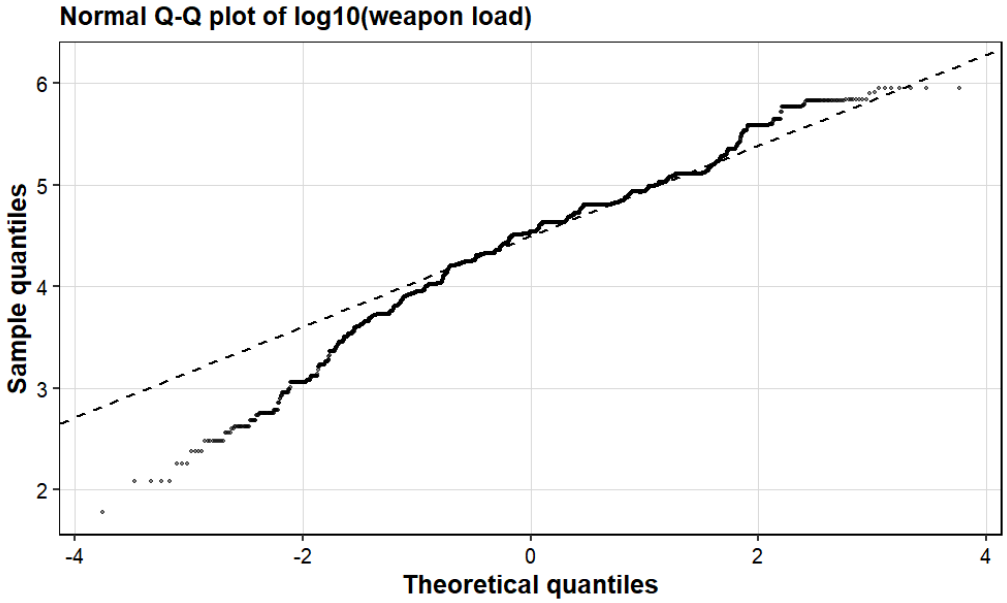


Figure 2.3: Normal Q-Q plot of log10(load): the middle follows the line, but both tails bend away.

Lognormal model: log10(load) has mean 4.4669 and sd 0.5616, i.e. meanlog = 10.2854 and sdlog = 1.2931 in natural logs. The goodness-of-fit chi-square statistic is 610.2 on 5 degrees of freedom ($p < 2.2e\text{-}16$): the lognormal model is far better than the normal (statistic 610 vs 7,338) but is still formally rejected, because the sample is large and the observed bin counts are not smooth (727, 582, 550, 653, 1,163, 1,006, 825, 388 against 736.8 expected in each bin). The middle bins contain more records and the tails fewer than the model predicts; the fitted quartiles (12,249 / 29,302 / 70,097) differ from the observed ones (15,930 / 34,260 / 63,720) in both tails.

Threshold load	Model probability P(load > threshold)	Observed share
25,000 lb	0.5489	0.5976
50,000 lb	0.3397	0.3522
75,000 lb	0.2337	0.1971
1,00,000 lb	0.1713	0.1318

Table 8: Probabilities from the fitted lognormal model against the observed shares. P(20,000 < load < 60,000) is 0.3264 by the model and 0.3453 observed.

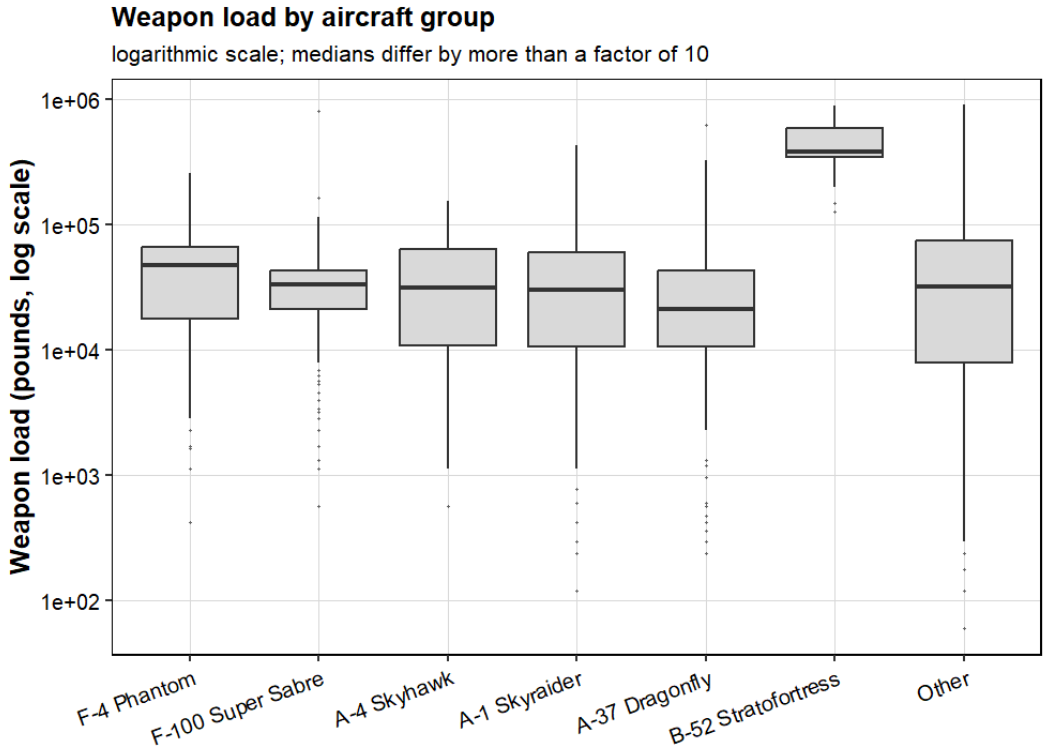


Figure 2.4: Weapon load by aircraft group on the log scale. The B-52 group (median 3,82,320 lb) is an extreme group, more than ten times the median load of the F-4 (47,790 lb) or the A-37 (21,240 lb).

Inference: the load distribution is a mixture of very different aircraft groups, not one single population. A single lognormal cannot fit a mixture, which explains the formal rejection even though the lognormal is qualitatively the right family. This is a limitation of the source data (the load is only recorded for one source database) rather than a failure of the method.

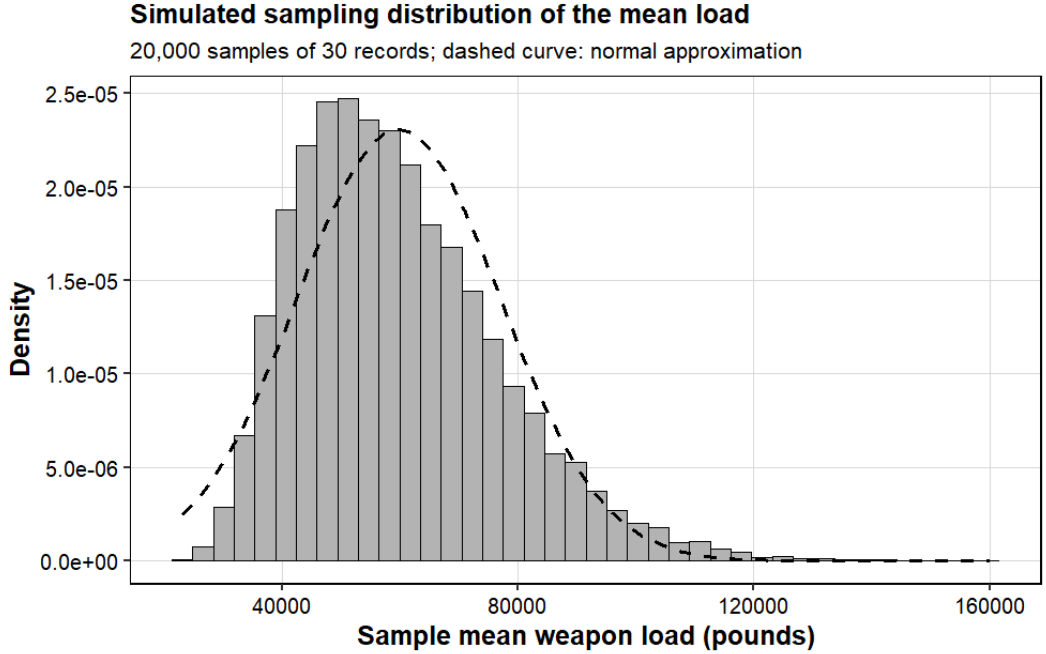


Figure 2.5: Simulated sampling distribution of the mean load of 30 records (20,000 repetitions) against the normal approximation.

Central limit theorem: for n = 30 the standard error of the mean is 94,711.3 / sqrt(30) = 17,291.8 lb. The simulation gives mean 59,736.9 lb (theory 59,795.3) and standard deviation 17,169.1 lb (theory 17,291.8) - both match well. The theoretical normal probability P(sample mean > 70,000) is 0.2775

while the simulation gives 0.2494; the distribution of the means is still slightly right-skewed (skewness 0.822), so n = 30 is not yet large for this very heavy-tailed variable.

Conclusion

The normal model is clearly wrong for the weapon load (impossible negative region, statistic 7,338). The lognormal model captures the shape well on the log scale and gives useful probabilities (for example about 34% of the records exceed 50,000 lb against 35% observed), but it is formally rejected by the chi-square test (610.2, $p < 2.2e-16$) because the data are a mixture of aircraft groups with very different loads. The central limit theorem was verified by simulation: the mean of 30 records has the theoretical mean and standard error, and the normal approximation for such means is adequate but not exact for this heavy-tailed variable.

Problem statement 3 (Unit III): Day versus night weapon load (two-sample t-test)

Problem statement

Test, at the 5% level, whether the mean weapon load of a delivery record differs between day and night missions. State the hypotheses, check the assumptions of the test, and cross-check the conclusion with a test that does not assume normality.

Introduction (purpose and approach)

Period of day is one of the few operational factors that is recorded for almost every record (PERIOD missing for only 4.2%). If night deliveries carry systematically different loads, mission planning and the interpretation of the earlier descriptive graphs must take that into account. The two-sample t-test compares the mean load between the day and night groups; because the load is extremely skewed, the test is applied to log10(load), which turns the comparison into a comparison of geometric means. Welch's version of the test is used because it does not assume equal variances.

Approach / algorithm with explanation

- H0: the mean log10(load) is the same for day and night records; H1: the means differ (two-sided), level of significance 5%.

1. Keep the 5,891 records with both a recorded load and a recorded period.
2. Check normality of log10(load) inside each group (skewness, Shapiro-Wilk on a subsample of 2,000) and equality of variances (F test and Levene's test).
3. Carry out Welch's two-sample t-test and report the difference of means on the log scale, its 95% confidence interval and the ratio of geometric means.
4. Measure the size of the effect with Cohen's d on the log scale.
5. Cross-check with the Mann-Whitney test on the raw loads, which assumes neither normality nor equal variances.
6. Investigate the apparent contradiction between the raw means and the medians.

Output with description

Group	n	Mean (lb)	sd (lb)	Median (lb)	Q1	Q3	Geometric mean (lb)
Day	4,196	57,614.7	80,064.1	42,240	17,600	65,600	32,254
Night	1,695	65,258.9	1,23,637.8	24,900	8,900	63,720	23,165

Table 9: Weapon load by period of day (5,891 records with both variables recorded; 3 records had a load but no valid period).

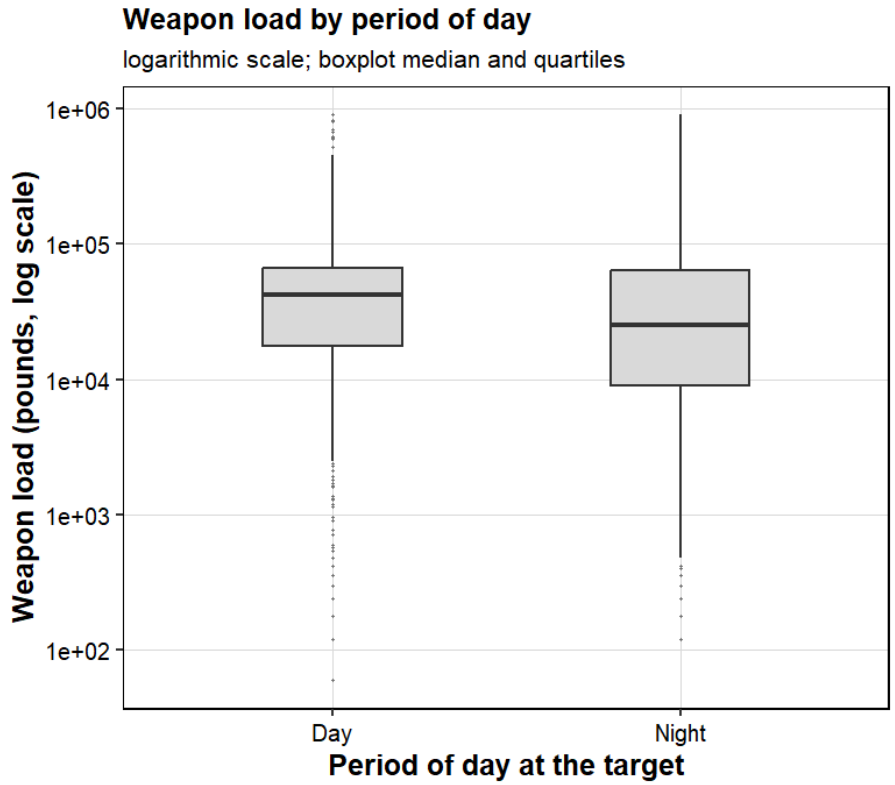


Figure 3.1: Weapon load (raw pounds) by period of day. The night group has a lower median but a longer upper tail.

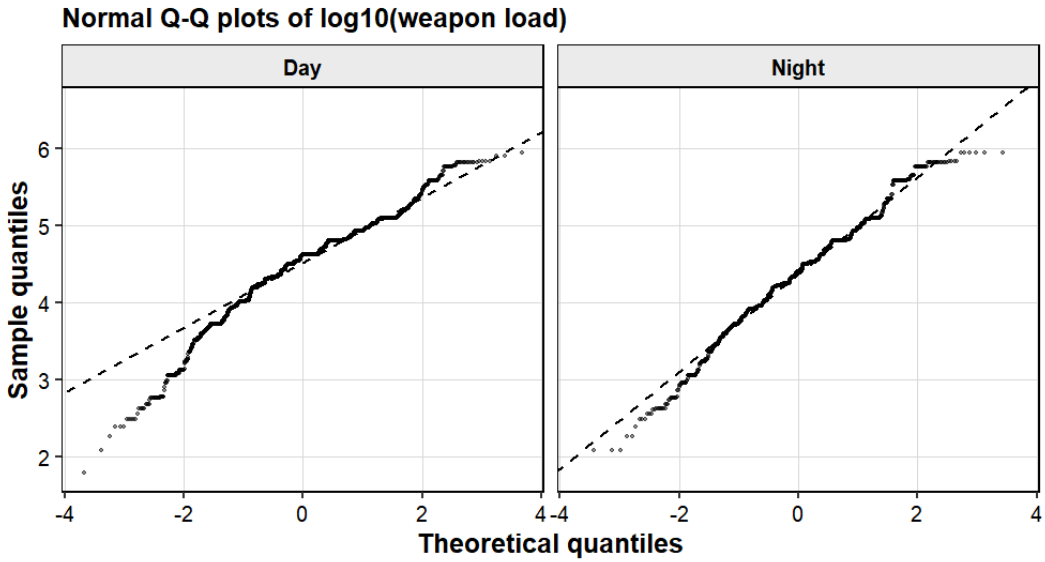


Figure 3.2: Normal Q-Q plots of log10(load) by period; both groups are close to normal on the log scale but the tails deviate.

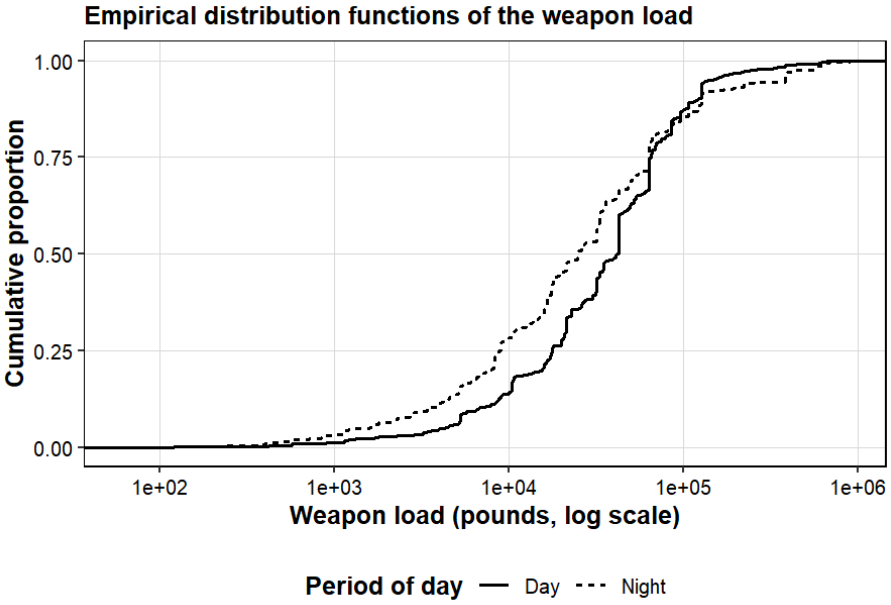


Figure 3.3: Empirical distribution functions of the weapon load (logarithmic axis) by period; the day curve lies to the right for most of the range.

Assumption checks: on the log scale the day group has skewness -0.79 and the night group -0.28 (against +4.55 for the raw load), so the log transformation makes the groups nearly symmetric; the Shapiro-Wilk test still rejects strict normality (day W = 0.9583, night W = 0.9876, both p < 1e-10) as it does for almost any large sample. The variance ratio test (F = 0.605) and Levene's test (F = 143.1, p < 2.2e-16) both indicate unequal variances, so Welch's version of the t-test is the correct one.

Welch two-sample t-test result (on log10(load)): t = 8.069 on 2,561.6 degrees of freedom, p-value = 1.079e-15, so H0 is rejected. The mean log10(load) is 4.5086 for day and 4.3648 for night; the difference (day - night) is 0.1438 with 95% confidence interval [0.1088, 0.1787]. On the original scale this means the day geometric mean is 1.392 times the night geometric mean (39.2% higher), with interval [1.285, 1.509], i.e. between 28.5% and 50.9% higher. Cohen's d on the log scale is 0.258, a small to moderate effect.

The raw group means tell the opposite story (night 65,258.9 lb against day 57,614.7 lb). The reason is visible in the data: night deliveries are polarised. 27.9% of night records deliver less than 10,000 lb (day: 13.8%) but 6.8% deliver more than 2,00,000 lb (day: 3.2%), so the night mean is pulled up by a small number of very heavy deliveries while the typical (median, geometric mean) night record is lighter. The median day load is 42,240 lb against 24,900 lb at night, a difference of 17,340 lb in the same direction as the t-test.

Mann-Whitney cross-check: U = 4,112,513, p < 2.2e-16; the raw-load medians differ significantly in the same direction. The conclusion does not depend on the normality assumption.

Conclusion

There is strong evidence (p = 1.08e-15) that day and night deliveries differ: the typical day load is about 39% heavier than the typical night load (geometric means 32,254 lb against 23,165 lb; medians 42,240 lb against 24,900 lb). The effect is small to moderate (d = 0.26). The raw means disagree only because a few very heavy night deliveries (large formations) pull the night mean up - a textbook example of why skewed data should be compared on the log scale or with medians. Limitation: records of the same mission are not strictly independent, so the p-value is slightly optimistic; the effect size and the confidence interval remain the reliable quantities.

Problem statement 4 (Unit III): Aircraft type and target country (chi-square test of independence)

Problem statement

Test whether the mix of aircraft types used in a delivery record is independent of the target country. Use the chi-square test of independence, measure the strength of the association and identify which combinations of aircraft and country occur more or less often than expected.

Introduction (purpose and approach)

Different aircraft have different roles (close air support, interdiction, strategic bombing); the question is whether the recorded aircraft mix actually changes from country to country, or whether the same mix is used everywhere. Both variables are categorical, so the chi-square test of independence is the appropriate test. The strength of the association is measured with Cramer's V and the individual cells are analysed with standardised residuals.

Approach / algorithm with explanation

- H0: aircraft type and target country are independent; H1: they are associated (level of significance 5%).

1. Keep the records with both variables recorded: 19,651 records. The 82 records with targets in Thailand are then excluded from the test because two expected cell counts fall below 5, which would break the chi-square approximation (Thailand is 0.4% of the sample); 19,569 records remain.
2. Build the contingency table (7 aircraft groups x 4 countries) and the row percentages.
3. Check the assumptions: each record is in exactly one cell, the minimum expected count is 40.96 (all above 5), and records are not fully independent (several records per mission - limitation stated once).
4. Chi-square test of independence; Cramer's V for the effect size.
5. Standardised residuals (observed - expected) / sqrt(expected) per cell; |residual| > 3 marks the cells that drive the association.

Output with description

Aircraft group	South Vietnam	North Vietnam	Laos	Cambodia	Total
F-4 Phantom	1,743 (36.5%)	784 (16.4%)	2,093 (43.9%)	149 (3.1%)	4,769
F-100 Super Sabre	2,516 (90.8%)	9 (0.3%)	140 (5.1%)	107 (3.9%)	2,772
A-4 Skyhawk	948 (51.5%)	542 (29.4%)	350 (19.0%)	1 (0.1%)	1,841
A-1 Skyraider	1,060 (51.7%)	94 (4.6%)	836 (40.8%)	59 (2.9%)	2,049
A-37 Dragonfly	1,222 (85.4%)	0 (0.0%)	22 (1.5%)	187 (13.1%)	1,431
B-52 Stratofortress	409 (49.0%)	67 (8.0%)	231 (27.7%)	128 (15.3%)	835
Other types	2,612 (44.5%)	867 (14.8%)	2,064 (35.1%)	329 (5.6%)	5,872
Total	10,510	2,363	5,736	960	19,569

Table 10: Contingency table of aircraft group against target country (row percentages in brackets).

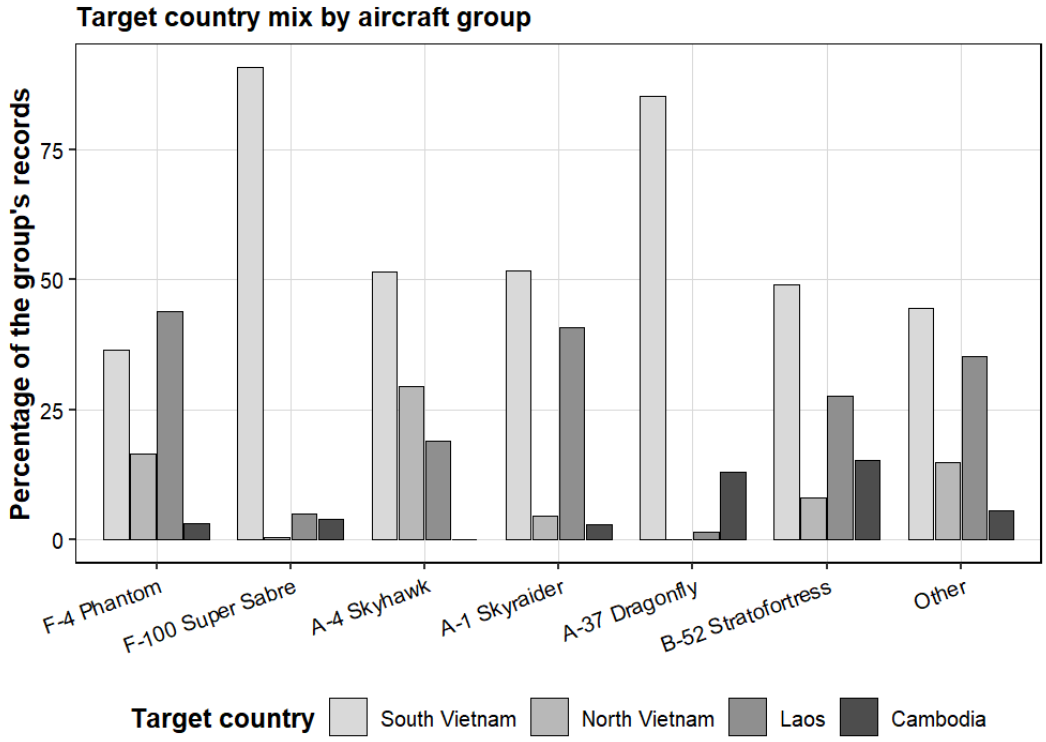


Figure 4.1: Target country mix (row percentages) by aircraft group. Aircraft types clearly differ in where they operate.

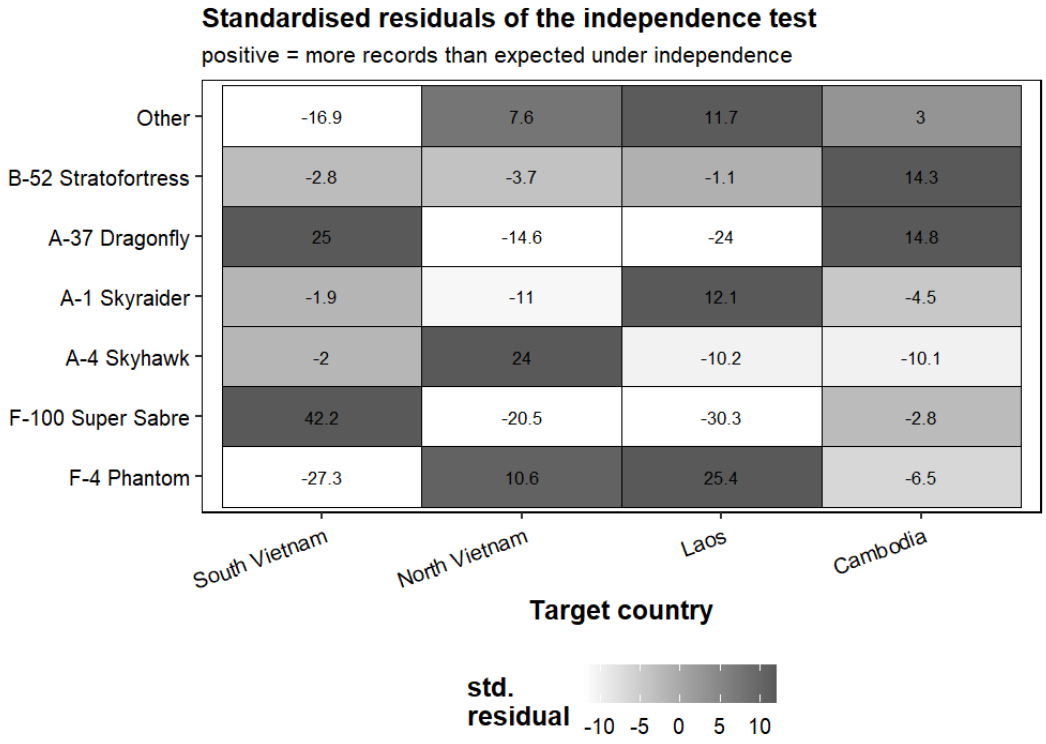


Figure 4.2: Standardised residuals of the test; darker cells deviate more from independence (positive = more records than expected).

Test result: chi-square = 4,538.3 on (7-1)(4-1) = 18 degrees of freedom, $p < 2.2e\text{-}16$, so H0 is rejected: the aircraft mix depends on the target country. Cramer's V = 0.278, a moderate association.

The largest standardised residuals identify how the association works: the F-100 and the A-37 operate almost only over South Vietnam (residuals +42.2 and +25.0; 90.8% and 85.4% of their records). The A-4 Skyhawk (a US Navy aircraft) is over represented over North Vietnam (+24.0;

29.4% of its records against 12.1% of the sample) and the F-4 Phantom is over represented over Laos (+25.4; 43.9%). The B-52 is over represented over Cambodia (+14.3; 15.3% of its records against 4.9%) - the well-known pattern of the secret Cambodian bombing. The F-4 is less used over South Vietnam than expected (-27.3) and the F-100 almost never over North Vietnam (-20.5). The overall conclusion is that the aircraft-country combination is a fairly strong role effect: each aircraft family has a characteristic theatre.

Conclusion

Aircraft type and target country are not independent (chi-square = 4,538.3, df = 18, $p < 2.2e-16$, Cramer's V = 0.278). Interpreting the residuals: fighter-bombers like the F-100 and A-37 fly close air support over South Vietnam, the F-4 covers Laos and the North, the carrier-based A-4 concentrates on North Vietnam, and the B-52 is heavily over represented over Cambodia. Note the limitation: records are not strictly independent (several per mission) and the Thailand cells had to be excluded to keep the chi-square approximation valid, so the p-value is optimistic; the effect size (moderate) and the residual pattern are the robust results.

Problem statement 5 (Unit IV): Explaining the size of the attacking formation (correlation and multiple regression)

Problem statement

Study the association between the numerical variables of the data and build a multiple linear regression model for the number of aircraft in a delivery record (log10 of AIRCRAFT_NO) as a function of aircraft type, target country, period of day, year and source database. Evaluate the model, check its assumptions and use it to predict the number of aircraft for a given mission profile.

Introduction (purpose and approach)

The number of aircraft in a record measures the size of the formation that flew the delivery. The purpose is to quantify how much of the size of a formation is explained by the factors recorded in the data, which combination of factors produces the largest formations, and whether the model can be used for prediction. The analysis has two stages: correlation analysis of the numerical variables (Spearman coefficients are the reliable ones because of the skew), and a multiple linear regression of log10(AIRCRAFT_NO) on the five categorical/numerical factors, with full diagnostics.

Approach / algorithm with explanation

1. Pearson and Spearman correlations between AIRCRAFT_NO, WEAPONS_DELIVERED, LOAD_LB, TGT_HOUR, LAT and LON on pairwise complete observations.
2. As a baseline, a simple linear regression of log10(AIRCRAFT_NO) on log10(WEAPONS_DELIVERED).
3. Multiple regression log10(AIRCRAFT_NO) ~ AIRCRAFT_GROUP + COUNTRY + PERIOD + YEAR + SOURCE_DB on the 18,811 records with all variables recorded (reference categories: F-4 Phantom, South Vietnam, Day, SEADAB).
4. Report the coefficient table with 95% confidence intervals, R squared, the overall F test and interpret the largest coefficients as percentage changes (factor 10^b).
5. Diagnostics: residual plots, Shapiro-Wilk and moments of the residuals, Cook's distance for influential observations, and the variance inflation of YEAR against all other predictors.
6. Prediction example with a 95% prediction interval and comparison with the observed records of the same profile.

Output with description

Correlation analysis (Spearman coefficients, all $p < 2.2e-16$ unless stated): the strongest association is between weapons delivered and weapon load (rho = 0.648); aircraft per record correlate with the load (0.288) and with weapons delivered (0.173); the hour at target has a weak negative association with aircraft (-0.217); latitude and longitude are negatively correlated (-0.280) simply because the theatre runs from north-west to south-east. The Pearson coefficients are much smaller for the heavy-tailed pairs (for example 0.247 against 0.173 for aircraft and weapons), confirming that the skew distorts the linear correlation.

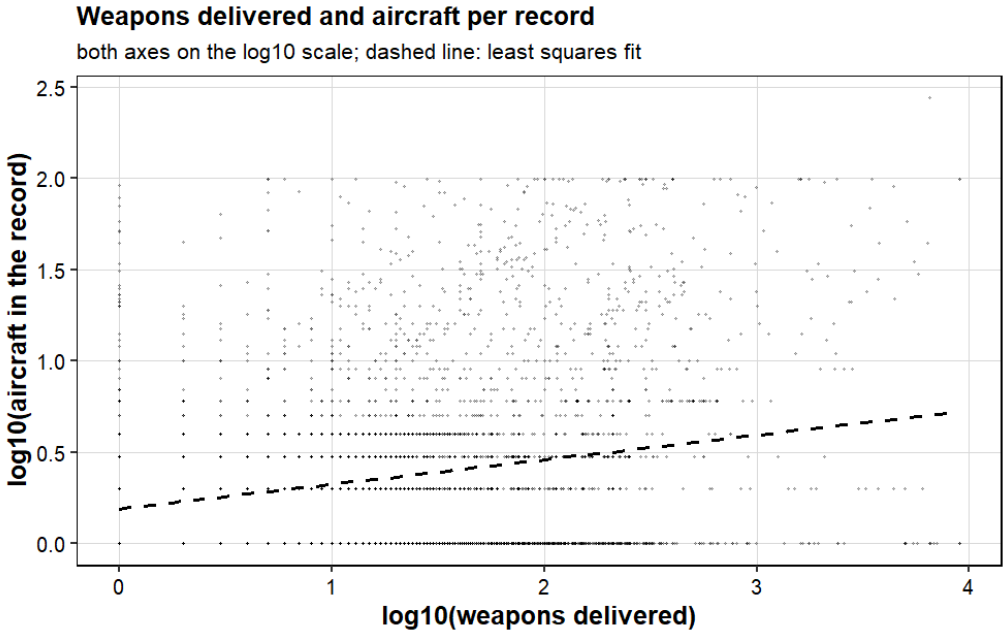


Figure 5.1: log10(weapons delivered) against log10(aircraft per record); the dashed line is the least squares fit of the simple regression, R squared = 0.063.

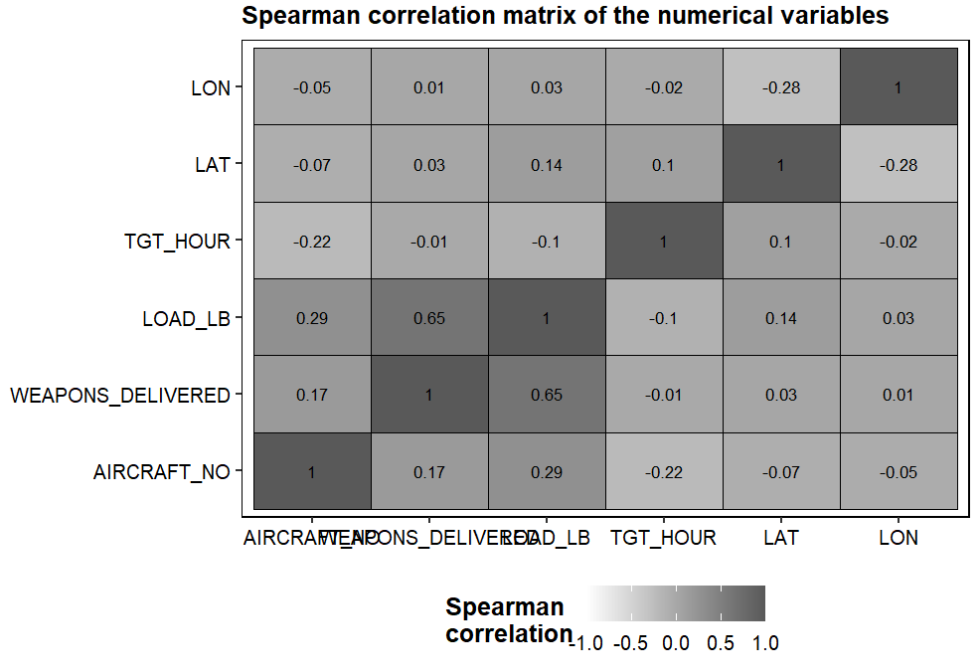


Figure 5.2: Spearman correlation matrix of the numerical variables.

Model	R squared	Adjusted R sq.	F / p	Residual std. error	AIC
Simple: aircraft on weapons	0.0634	0.0633	1,273 / < 2.2e-16	0.2594	2,623.8
Multiple: all five factors	0.1660	0.1654	267.2 / < 2.2e-16	0.2449	467.1

Table 11: Comparison of the simple and the multiple regression (n = 18,811).

Multiple regression result: R squared = 0.166 (adjusted 0.1654), so the five factors together explain about 17% of the variation of log10(aircraft) between records; the overall F test is 267.2 on (14, 18,796) degrees of freedom, p < 2.2e-16. The largest effects on the size of a formation, holding the other factors constant:

Predictor (against reference)	Estimate (log10)	Effect on aircraft	95% CI (log10)

B-52 Stratofortress (vs F-4)	+0.193	+55.9% (x1.56)	[0.164, 0.221]
Period: Night (vs Day)	-0.200	-36.9%	[-0.208, -0.192]
Source CACTA6669 (vs SEADAB)	-0.189	-35.3%	[-0.203, -0.175]
Source CACTA6570 (vs SEADAB)	-0.103	-21.1%	[-0.117, -0.089]
Country: Cambodia (vs South Vietnam)	-0.087	-18.1%	[-0.105, -0.070]
A-37 Dragonfly (vs F-4)	-0.074	-15.6%	[-0.089, -0.058]
Year (per year)	-0.033	-7.3% per year	[-0.036, -0.030]

Table 12: Largest coefficients of the multiple regression (all $p < 0.001$). The Thailand coefficient (+0.379, +139%) is significant but rests on only 82 records, so it is not interpreted. F-100 and 'Other' are not significantly different from the F-4 at the 5% level once the other factors are held constant; the A-1 and A-4 coefficients are small but significant (-6.1% and -5.2%).

Interpretation: B-52 records involve about 1.56 times as many aircraft as the equivalent F-4 records - the classic multi-aircraft cell formations. Night records are about 37% smaller than day records of the same type and country, and the two smaller source databases (CACTA) record formations about 35% and 21% smaller than SEADAB, which is consistent with them recording different kinds of missions. Formations shrink by about 7% per war year, and attacks recorded over Cambodia are about 18% smaller than over South Vietnam.

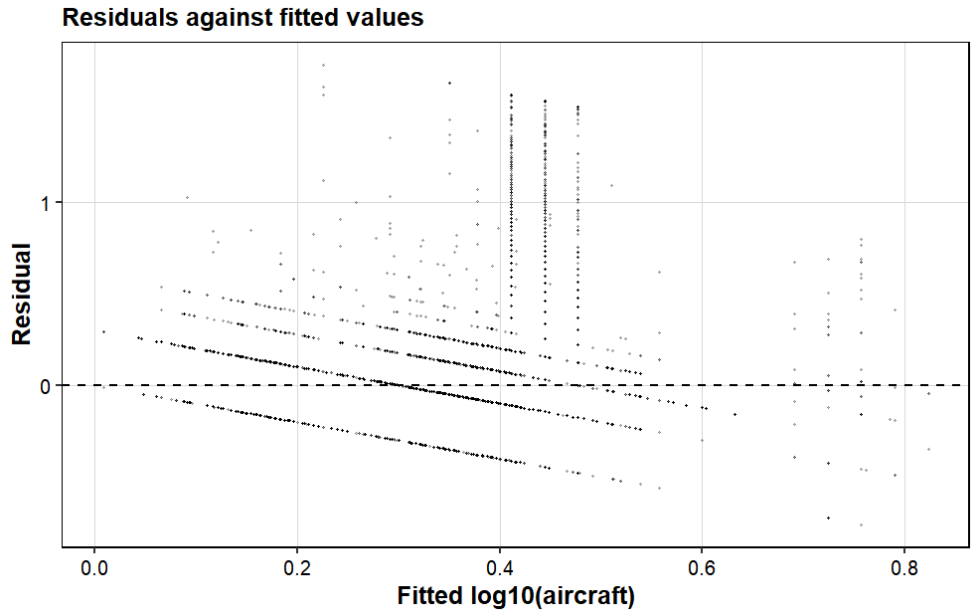


Figure 5.3: Residuals against fitted values. The diagonal stripes come from the discrete response values (log10(1) = 0, log10(2) = 0.30, ...), not from a modelling problem; the points scatter evenly around zero.

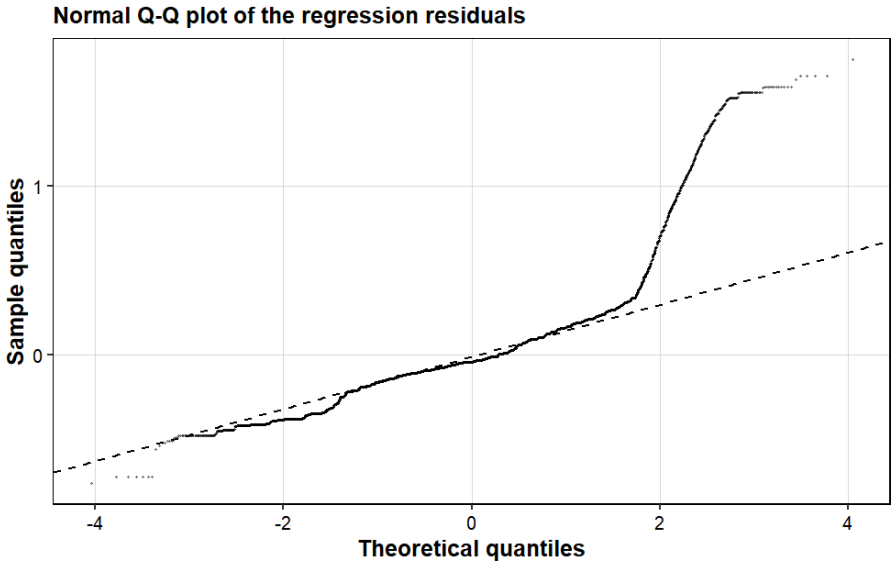


Figure 5.4: Normal Q-Q plot of the residuals; the model is adequate in the middle, the right tail corresponds to a few very large formations (up to 99 aircraft) that the model under predicts.

Model checks: the residuals have mean zero by construction; skewness 2.45 and kurtosis 11.4 (Shapiro-Wilk W = 0.822 on a subsample of 2,000, $p < 2.2e-16$), driven by the right tail of very large formations, not by an error of specification. No observation is influential: the largest Cook's distance is 0.0089, far below 1, and only 2.81% of the records exceed the 4/n guideline. Collinearity is moderate: the variance inflation of YEAR against all other predictors is 3.76, below the usual threshold of 5, but part of the year trend overlaps with the country mix (for example Thailand targets appear only late in the war), so the year coefficient should be read as an average trend, not as a causal effect.

Prediction example: for the profile 'F-4 Phantom, target North Vietnam, night, 1972, SEADAB' the model predicts 1.55 aircraft with a 95% prediction interval of [0.51, 4.69]. The 32 observed records with exactly this profile have a mean of 1.75 aircraft, comfortably inside the interval. The interval is wide on the original scale (a factor of nine) - honest evidence that the model predicts the average behaviour of a group of records well, but a single record only roughly.

Conclusion

The size of a formation is dominated by the aircraft type (B-52 cells carry about 56% more aircraft than the F-4 reference) and by the period of day (night records are about 37% smaller). The five factors together explain about 17% of the record-to-record variation, which is realistic: many determinants of a formation (target importance, availability, weather, operation) are not in the data. The model passes the practical checks (no influential observations, moderate collinearity, sensible residuals) and its prediction interval covers the observed mean of a specific mission profile. Correlation results confirm the relationships and warn that Pearson coefficients on skewed variables understate the association (weapons-load: Pearson 0.31 against Spearman 0.65).

Overall conclusion

1. The data preparation pipeline (chunked reading of the 46-lakh-record file, weapon-delivery population definition, random sample of 20,000 records, cleaning with documented rules) produced a reproducible analysis file; every number in this report is regenerated by one command per script and logged in outputs/.

2. Problem 1: the recorded campaign peaks in 1970, concentrates on South Vietnam and Laos, is flown mainly by F-4, F-100, A-1, A-4 and A-37 aircraft and delivers mostly bombs; every numerical variable is strongly right-skewed, so the typical record is described by medians and quartiles rather than means.

3. Problem 2: the weapon load is lognormal in shape (far better than the normal model) but the fit is formally rejected because the load is a mixture of very different aircraft groups (B-52 median 3.82 lakh lb against A-37 21,240 lb); model-based probabilities are still accurate to within about 4 percentage points; the central limit theorem check for the mean of 30 records matched theory in mean and standard error.

4. Problem 3: day deliveries carry about 39% more load than night deliveries (geometric means; medians 42,240 against 24,900 lb; Welch p = 1.08e-15, d = 0.26) - the raw means mislead because a few very heavy night formations pull the night mean up.

5. Problem 4: the aircraft mix is not independent of the target country (chi-square = 4,538.3, df = 18, p < 2.2e-16, Cramer's V = 0.278); the standardised residuals show exactly how each aircraft family is tied to a theatre.

6. Problem 5: log10 of the aircraft per record is explained by type, period, country, year and source (R squared = 0.166, F = 267.2); B-52 records carry about 1.56 times the aircraft of an F-4 record and night records about 37% fewer; the prediction example validated against the observed records.

7. Across the five problems one methodological lesson repeats: the recorded quantities (weapons, load) are extremely right-skewed, so medians, geometric means and log-scale tests describe the data honestly, while raw means are distorted by a small number of very large deliveries.

8. Main limitations, stated consistently: records are delivery events (not independent missions), the load is recorded only in SEADAB, several raw columns are unusable (WEAPONTYPECLASS 100% empty, BDA 94% empty), and the source file mixes four databases without de-duplication flags.

R packages and software used

R 4.6.1 with RStudio-compatible plain scripts. Packages: readr (chunked reading of the large raw file), ggplot2 (all figures, black-and-white theme), base R statistics (lm, t.test, chisq.test, shapiro.test, wilcox.test). The Levene test, the moment-based skewness/kurtosis and the variance inflation calculation are implemented in base R in scripts/common_setup.R so that the project runs with the standard library only. File map: scripts/00_profile_raw.R and 00b/00c (profiling the raw file), scripts/01_clean.R (builds clean_data.csv), scripts/problem1.R to problem5.R (one script per problem statement), outputs/ (text logs used in this report), figures/ (all graphs).